

Q1.

Let X be a continuous random variable with probability density function given by

$$f(x) = \begin{cases} \frac{3}{4}x(2-x) & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Find $P(X = 1)$

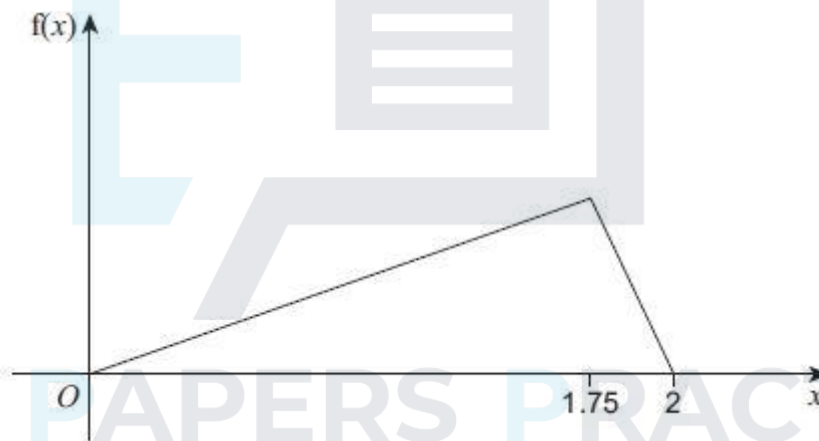
Circle your answer.

0 $\frac{1}{2}$ $\frac{3}{4}$ $\frac{27}{32}$

(Total 1 mark)

Q2.

The diagram shows a graph of the probability density function of the random variable X .



(a) State the mode of X .

(1)

(b) Find the probability density function of X .

(4)

(Total 5 marks)

Q3.

The random variable T has probability density defined by

$$f(t) = \begin{cases} \frac{t}{8} & 0 \leq t \leq k \\ 0 & \text{otherwise} \end{cases}$$

Find the value of k

Circle your answer.

$$\frac{1}{16}$$

$$\frac{1}{4}$$

$$4$$

$$16$$

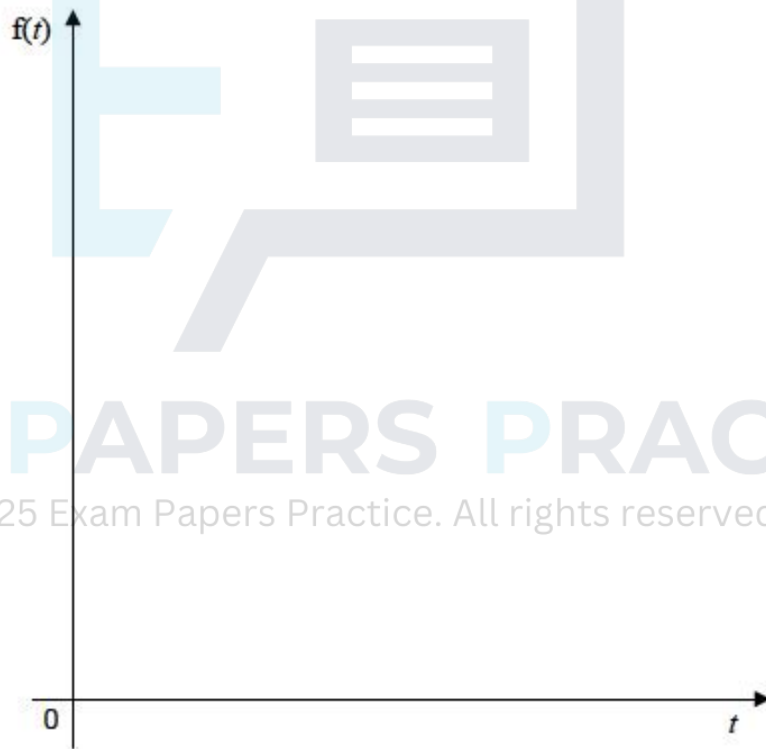
(Total 1 mark)

Q4.

The continuous random variable T has probability density function defined by

$$f(t) = \begin{cases} \frac{1}{3} & 0 \leq t \leq \frac{3}{2} \\ \frac{9-2t}{18} & \frac{3}{2} \leq t \leq \frac{9}{2} \\ 0 & \text{otherwise} \end{cases}$$

- (a) (i) Sketch this probability density function below.



(2)

- (ii) State the median of T .

(1)

- (b) (i) Find $E(T)$

(2)

- (ii) Given that $E(T^2) = \frac{15}{4}$, find $\text{Var}(4T - 5)$

(3)

Q5.

A continuous random variable X has probability density function defined by

$$f(x) = \begin{cases} kx^2 & 0 \leq x \leq 3 \\ 9k & 3 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(3)

- (b) Show that the value of k is $\frac{1}{18}$.

(4)

- (c) (i) Write down the median value of X .
(ii) Calculate the value of the lower quartile of X .

(4)

(Total 11 marks)

Q6.

The time, in weeks, that a patient must wait to be given an appointment in Holmsoon Hospital may be modelled by a random variable T having the cumulative distribution function

$$F(t) = \begin{cases} 0 & t < 0 \\ \frac{t^3}{216} & 0 \leq t \leq 6 \\ 1 & t > 6 \end{cases}$$

- (a) Find, to the nearest day, the time within which 90 per cent of patients will have been given an appointment.

(3)

- (b) Find the probability density function of T for all values of t .

(3)

- (c) Calculate the mean and the variance of T .

(6)

- (d) Calculate the probability that the time that a patient must wait to be given an appointment is more than one standard deviation above the mean.

(4)

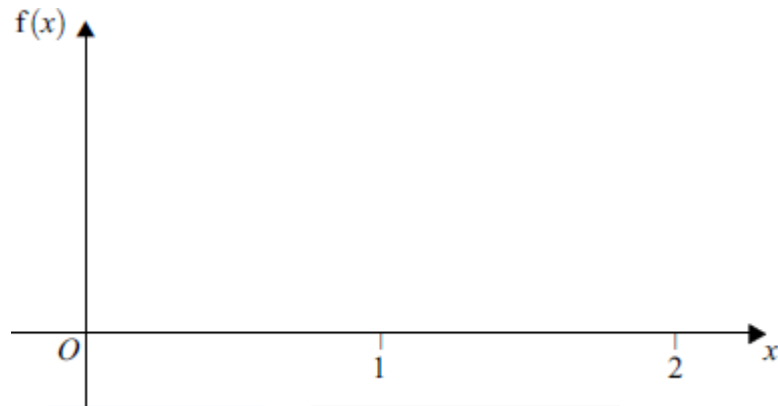
(Total 16 marks)

Q7.

A continuous random variable X has the probability density function defined by

$$f(x) = \begin{cases} x^2 & 0 \leq x \leq 1 \\ \frac{1}{3}(5 - 2x) & 1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f on the axes below.



(3)

- (b) (i) Find the cumulative distribution function, F , for $0 \leq x \leq 1$.

(2)

- (ii) Hence, or otherwise, find the value of the lower quartile of X .

(2)

- (c) (i) Show that the cumulative distribution function for $1 \leq x \leq 2$ is defined by

$$F(x) = \frac{1}{3} (5x - x^2 - 3)$$

(4)

- (ii) Hence, or otherwise, find the value of the upper quartile of X .

(4)

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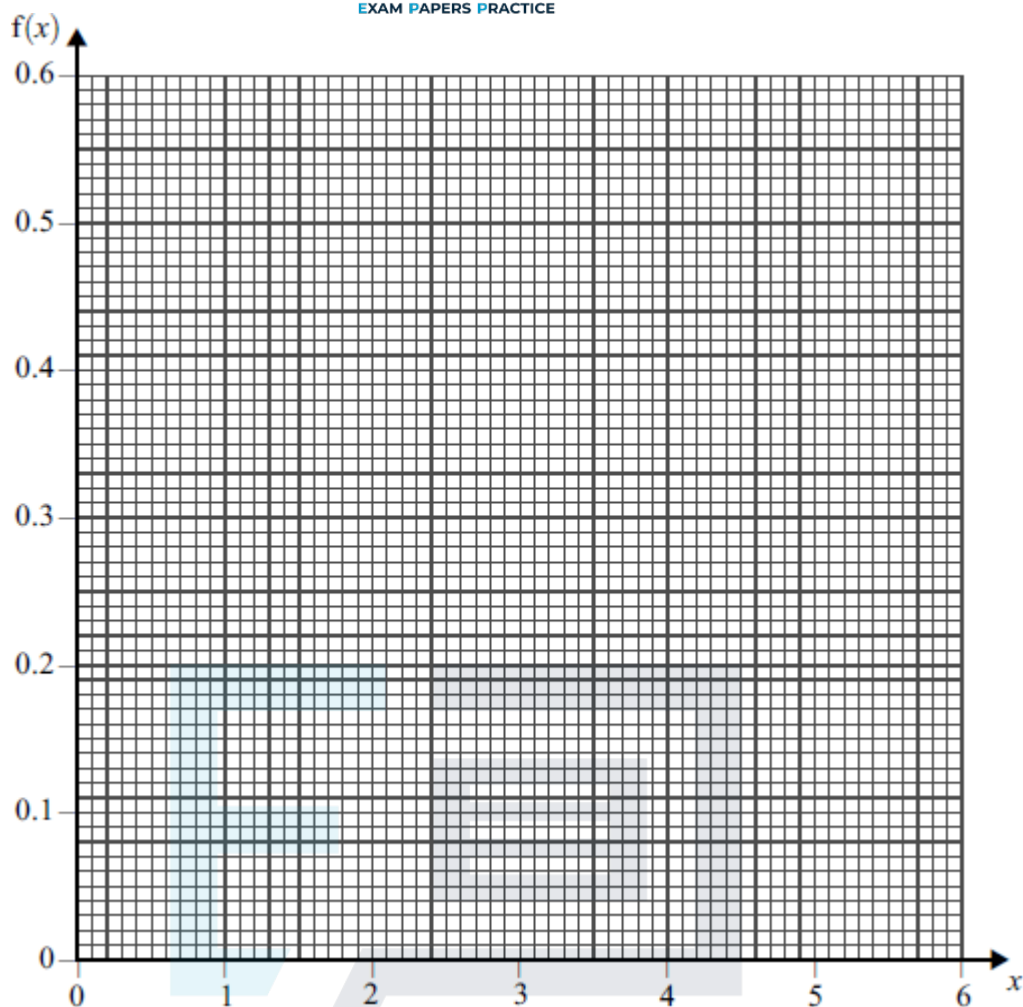
(Total 15 marks)

Q8.

A continuous random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{1}{6}(4 - x) & 1 \leq x \leq 3 \\ \frac{1}{6} & 3 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Draw the graph of f on the grid below.



(2)

- (b) Prove that the mean of X is $2\frac{5}{9}$.

(4)

- (c) Calculate the **exact** value of:

(i) $P(X > 2.5)$;

(2)

(ii) $P(1.5 < X < 4.5)$;

(3)

(iii) $P(X > 2.5 \text{ and } 1.5 < X < 4.5)$;

(2)

(iv) $P(X > 2.5 \mid 1.5 < X < 4.5)$.

(2)

(Total 15 marks)

Q9.

The continuous random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{3}{8}x^2 & 0 \leq x \leq \frac{1}{2} \\ \frac{3}{32} & \frac{1}{2} \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f . (3)
- (b) Show that:
- (i) $P(X \geq \frac{1}{3}) = \frac{1}{4}$
- (ii) $P(X \geq \frac{3}{4}) = \frac{3}{4}$ (3)
- (c) Hence write down the **exact** value of:
- (i) the interquartile range of X ;
- (ii) the median, m , of X . (3)
- (d) Find the **exact** value of $P(X < m \mid X \geq \frac{3}{4})$. (3)
- (Total 12 marks)

Q10.

The error, in minutes, made by Paul in estimating the time that he takes to complete a college assignment may be modelled by the random variable T with probability density function

$$f(t) = \begin{cases} \frac{1}{30} & -5 \leq t \leq 25 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find:
- (i) $E(T)$; (1)
- (ii) $\text{Var}(T)$. (1)
- (b) Calculate the probability that Paul will make an error of magnitude at least 2 minutes when estimating the time that he takes to complete a given assignment. (3)

Q11.

The continuous random variable X has probability density function given by

$$f(x) = \begin{cases} \frac{1}{2}(x^2 + 1) & 0 \leq x \leq 1 \\ (x - 2)^2 & 1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f . (3)
- (b) Calculate $P(X \leq 1)$. (3)
- (c) Show that $E(X^2) = \frac{4}{5}$. (5)
- (d) (i) Given that $E(X) = \frac{19}{24}$ and that $\text{Var}(X) = \frac{499}{k}$, find the numerical value of k . (3)
- (ii) Find $E(5X^2 + 24X - 3)$. (2)
- (iii) Find $\text{Var}(12X - 5)$. (2)

(Total 18 marks)

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Q12.

The random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{1}{2} & 0 \leq x \leq 1 \\ \frac{1}{8}(x - 4)^2 & 1 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) State values for the median and the lower quartile of X . (2)
- (b) Show that, for $1 \leq x \leq 4$, the cumulative distribution function, $F(x)$, of X is given by

$$F(x) = 1 + \frac{1}{54}(x-4)^3$$

(You may assume that $\int (x-4)^2 dx = \frac{1}{3}(x-4)^3 + c$.)

(4)

- (c) Determine $P(2 \leq X \leq 3)$.

(2)

- (d) (i) Show that q , the upper quartile of X , satisfies the equation $(q-4)^3 = -13.5$.

(3)

- (ii) Hence evaluate q to three decimal places.

(1)

(Total 12 marks)

Q13.

The continuous random variable X has the probability density function given by

$$f(x) = \begin{cases} \frac{1}{16}x^3 & 0 \leq x \leq 2 \\ \frac{1}{6}(5-x) & 2 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(3)

- (b) Prove that the cumulative distribution function of X for $2 \leq x \leq 5$ can be written in the form

$$F(x) = 1 - \frac{1}{12}(5-x)^2$$

(4)

- (c) Hence, or otherwise, determine $P(X \geq 3 \mid X \leq 4)$.

(5)

(Total 12 marks)

Q14.

The continuous random variable X has probability density function given by



$$f(x) = \begin{cases} \frac{1}{2} & 0 \leq x \leq 1 \\ \frac{3-x}{4} & 1 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f . (3)
- (b) Explain why the value of η , the median of X , is 1. (2)
- (c) Show that the value of μ , the mean of X , is $\frac{13}{12}$. (4)
- (d) Find $P(X < 3\mu - \eta)$. (3)
- (Total 12 marks)

Q15.

The waiting time, X minutes, for fans to gain entrance to see an event may be modelled by a continuous random variable having the distribution function defined by

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{2}x & 0 \leq x \leq 1 \\ \frac{1}{54}(x^3 - 12x^2 + 48x - 10) & 1 \leq x \leq 4 \\ 1 & x > 4 \end{cases}$$

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- (a) (i) Sketch the graph of F . (4)
- (ii) Explain why the value of q_1 , the lower quartile of X , is $\frac{1}{2}$. (2)
- (iii) Show that the upper quartile, q_3 , satisfies $1.6 < q_3 < 1.7$. (3)
- (b) The probability density function of X is defined by

$$f(x) = \begin{cases} \alpha & 0 \leq x \leq 1 \\ \beta(x-4)^2 & 1 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (i) Show that the **exact** values of α and β are $\frac{1}{2}$ and $\frac{1}{18}$ respectively. (5)
- (ii) Hence calculate $E(X)$. (5)
- (Total 19 marks)**

Q16.

The delay, in hours, of certain flights from Australia may be modelled by the continuous random variable T , with probability density function

$$f(t) = \begin{cases} \frac{2}{15}t & 0 \leq t \leq 3 \\ 1 - \frac{1}{5}t & 3 \leq t \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f . (3)
- (b) Calculate:
- (i) $P(T \leq 2)$; (2)
- (ii) $P(2 < T < 4)$. (3)
- (c) Determine $E(T)$. (4)

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(Total 12 marks)

Q17.

The continuous random variable X has the probability density function given by

$$f(x) = \begin{cases} 3x^2 & 0 < x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Determine:
- (i) $E\left(\frac{1}{X}\right)$; (3)
- (ii) $\text{Var}\left(\frac{1}{X}\right)$.

(4)

- (b) Hence, or otherwise, find the mean and the variance of $\left(\frac{5+2X}{X}\right)$.

(5)

(Total 12 marks)

Q18.

Engineering work on the railway network causes an increase in the journey time of commuters travelling into work each morning.

The increase in journey time, T hours, is modelled by a continuous random variable with probability density function

$$f(t) = \begin{cases} 4t(1-t^2) & 0 \leq t \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Show that $E(T) = \frac{8}{15}$. (3)

- (b) (i) Find the cumulative distribution function, $F(t)$, for $0 \leq t \leq 1$. (2)

- (ii) Hence, or otherwise, for a commuter selected at random, find
 $P(\text{mean} < T < \text{median})$

(5)

(Total 10 marks)

Q19.

The error, R centilitres, made in measuring the contents of bottles of wine can be modelled by the rectangular distribution

$$f(r) = \begin{cases} \alpha & -0.4 < r < 0.4 \\ 0 & \text{otherwise} \end{cases}$$

where α is a positive constant.

- (a) Find the value of α . (1)

- (b) Calculate the mean and the standard deviation of R . (3)

- (c) For a bottle of wine selected at random, calculate the probability that the **magnitude** of the error made in measuring its contents is less than 0.3 cl. (2)

(Total 6 marks)

Q20.

The continuous random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{1}{5}(2x + 1) & 0 \leq x \leq 1 \\ \frac{1}{15}(4 - x)^2 & 1 < x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(2)

- (b) (i) Show that the cumulative distribution function, $F(x)$, for $0 \leq x \leq 1$ is

$$F(x) = \frac{1}{5}x(x + 1)$$

(3)

- (ii) Hence find $P(X \geq 0.5)$.

(2)

- (iii) Verify that the lower quartile, q_1 , satisfies

$$0.5 < q_1 < 0.75$$

(3)

(Total 10 marks)

Q21.

The continuous random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{1}{5}(2x + 1) & 0 \leq x \leq 1 \\ \frac{1}{15}(4 - x)^2 & 1 < x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(2)

- (b) (i) Show that the cumulative distribution function, $F(x)$, for $0 \leq x \leq 1$ is

$$F(x) = \frac{1}{5}x(x + 1)$$

(3)

- (ii) Hence write down the value of $P(X \leq 1)$.

(1)

- (iii) Find the value of x for which $P(X \geq x) = \frac{17}{20}$.

(5)

(iv) Find the lower quartile of the distribution.

(4)

(Total 15 marks)

Q22.

The time, T hours, that the supporters of Bracken Football Club have to queue in order to obtain their Cup Final tickets has the following probability density function.

$$f(t) = \begin{cases} \frac{1}{5} & 0 \leq t < 3 \\ \frac{1}{45}t(6-t) & 3 \leq t \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

(a) Sketch the graph of f .

(3)

(b) Write down the value of $P(T = 3)$.

(1)

(c) Find the probability that a randomly selected supporter has to queue for at least 3 hours in order to obtain tickets.

(2)

(d) Show that the median queuing time is 2.5 hours.

(2)

(e) Calculate $P(\text{median} < T < \text{mean})$.

(6)

(Total 14 marks)

Q23.

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The error, X millimetres, made when the heights of prospective members of a new gym club are measured can be modelled by a rectangular distribution with the following probability density function.

$$f(x) = \begin{cases} k & -4 \leq x \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

(a) State the value of k .

(1)

(b) Write down the value of $E(X)$.

(1)

(c) Calculate $P(X > 0)$.

(2)

(d) The height of a randomly selected prospective member is measured. Find the

probability that the **magnitude** of the error made exceeds 3.5 millimetres.

(3)

(Total 7 marks)

Q24.

A machine cuts lengths of water hose for washing machines. Each length cut is at least 2 metres. The extra length, X millimetres, above 2 metres maybe modelled by the following probability density function, where c is a positive constant.

$$f(x) = \begin{cases} c & 0 \leq x \leq 4 \\ \frac{c}{16}(20 - x) & 4 \leq x \leq 20 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f . (4)
- (b) By considering your sketch, or otherwise, find the exact value of c . (3)
- (c) Hence determine the probability that a randomly selected length of hose is less than 2.01 metres. (4)
- (Total 11 marks)

Q25.

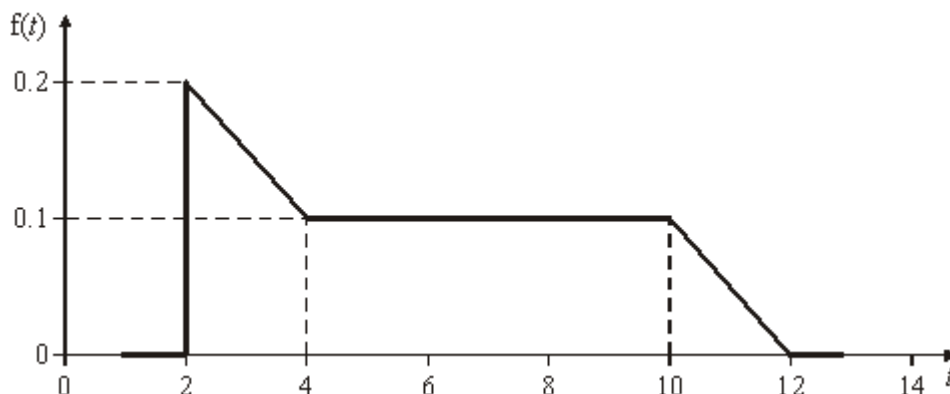
The continuous random variable X has the following cumulative distribution function.

$$f(x) = \begin{cases} 0 & x < 0 \\ \frac{x^3}{64} & 0 \leq x \leq 4 \\ 1 & x > 4 \end{cases}$$

- (a) Calculate $P(2 \leq X \leq 3)$. (2)
- (b) Find the probability density function, $f(x)$, for $0 \leq x \leq 4$. (2)
- (c) Evaluate:
- (i) $E(X)$; (3)
- (ii) $\text{Var}(X)$. (3)
- (Total 10 marks)

Q26.

The probability density function, $f(t)$, for the time, T minutes, taken by five-year-old children to complete a puzzle is shown by the following graph.



- (a) Determine:
- (i) $P(T < 10)$;
 - (ii) $P(T > 5)$.
- (3)
- (b) Hence determine the probability that a randomly selected five-year-old child completes the puzzle in less than 10 minutes, given that the child has not completed the puzzle within 5 minutes.
- (4)
- (Total 7 marks)

Q27.

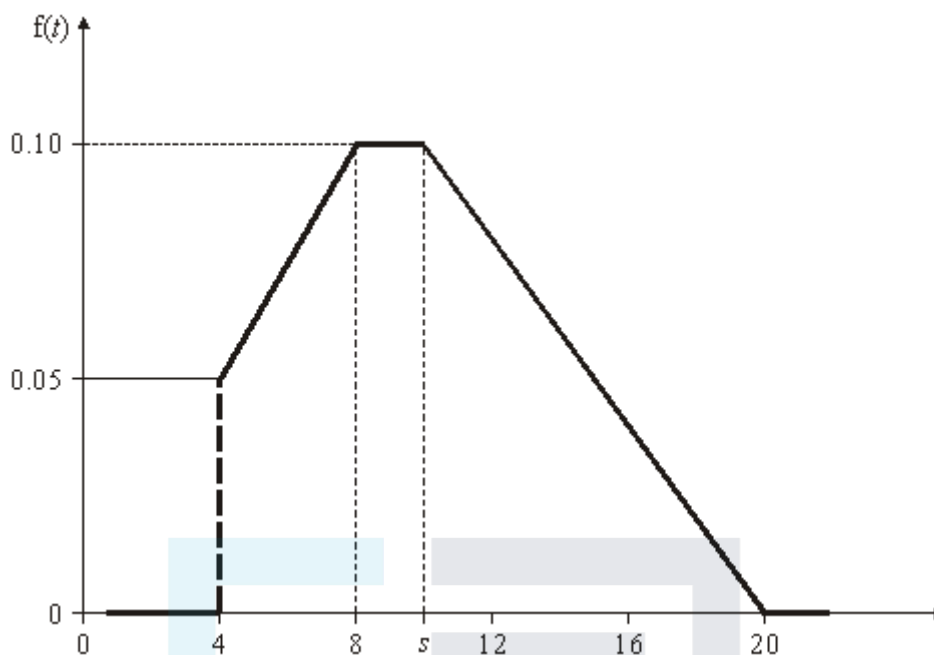
The time, T minutes, that Suneil waits to access a telephone banking service may be modelled by the following probability density function.

$$f(t) = \begin{cases} \frac{6}{19}t & 0 \leq t \leq 2 \\ \frac{6}{19}t(3-t) & 2 \leq t \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of $f(t)$.
- (2)
- (b) Show that the probability that Suneil waits less than 30 seconds is 0.0395, correct to three significant figures.
- (2)
- (c) Calculate Suneil's mean waiting time.
- (5)
- (Total 9 marks)

Q28.

At a hairdressers, the time, T minutes, required to attend to a male customer may be modelled by the probability density function, $f(t)$, as represented by the graph below.



- (a) Determine $P(T < 8)$ (2)
- (b) (i) Show why the value of s , marked on the graph above, must be 10. (3)
- (ii) Hence determine $P(T > 15)$. (3)
- (Total 8 marks)

Q29.

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The time, T hours, taken by any member of a group of friends to complete a run for charity can be modelled by the following probability density function.

$$f(t) = \begin{cases} \frac{1}{90}t^2 & 3 \leq t \leq 6 \\ 2 - \frac{4}{15}t & 6 \leq t \leq 7.5 \\ 0 & \text{otherwise} \end{cases}$$

- (a) (i) Show that the probability that a member of the group selected at random takes at least 6 hours to complete the run is 0.3. (2)
- (ii) Evaluate the median time taken to complete the run. (6)
- (b) Calculate the mean time taken to complete the run.

(5)
(Total 13 marks)

Q30.

A town's library offers a booking system for free internet access. Users are allowed a maximum of one hour of connection time at any one booking.

The actual connection times, X minutes, of users may be modelled by the following probability density function, where k is a constant.

$$f(x) = \begin{cases} \frac{kx}{40} & 0 \leq x \leq 40 \\ k & 40 \leq x \leq 60 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f . (4)
- (b) By considering your sketch, or otherwise, show that the value of k is 0.025. (3)
- (c) Hence determine $P(X > 30)$. (3)

(Total 10 marks)

Q31.

A continuous random variable T has the probability density function $f(t)$ given by:

$$f(t) = \begin{cases} \frac{1}{18}t^2 & 0 \leq t \leq 3 \\ \frac{1}{4}(5-t) & 3 \leq t \leq 5 \\ 0 & \text{otherwise.} \end{cases}$$

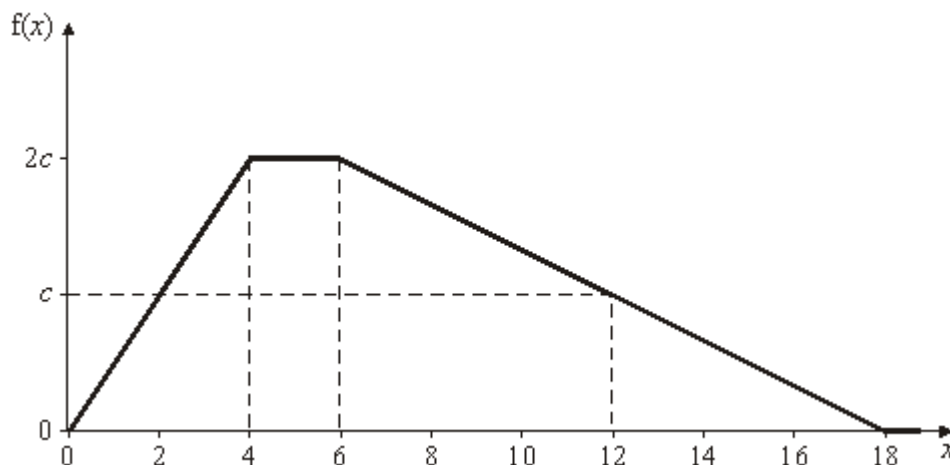
- (a) Sketch the graph of f . (2)
- (b) Determine the distribution function, $F(t)$, for $0 \leq t \leq 5$. (4)
- (c) Hence, or otherwise, calculate $P(T < 4)$. (2)

(Total 8 marks)

Q32.

The time, X minutes, by which an appointment is delayed at a health centre may be

modelled by a probability density function represented by the graph below.



- (a) Show that the value of the constant c , as used on the vertical axis, is 0.05. (3)
- (b) Determine the probability that an appointment is delayed:
- (i) by more than 4 minutes; (2)
 - (ii) by between 4 and 12 minutes; (3)
 - (iii) by less than 12 minutes, given that it is delayed by more than 4 minutes. (4)
- (c) Give a reason why the probability density function, as represented by the graph above, is unlikely to provide a satisfactory model for **all** delay times at this health centre. (1)

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(Total 13 marks)

Q33.

The time, in minutes, that vehicles have to wait at a T-junction may be represented by a random variable X with probability density function:

$$f(x) = \begin{cases} k(2.5 - x) & 0 \leq x \leq 2.5 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Sketch the graph of f . (3)
- (b) By considering your sketch. or otherwise, show that the value of the constant k is 0.32. (3)
- (c) Determine the probability that a vehicle has to wait at the T-junction for between 1 minute and 2 minutes. (3)

Q34.

At a Post Office, the time, T minutes, that customers have to wait in order to be served has the following probability density function.

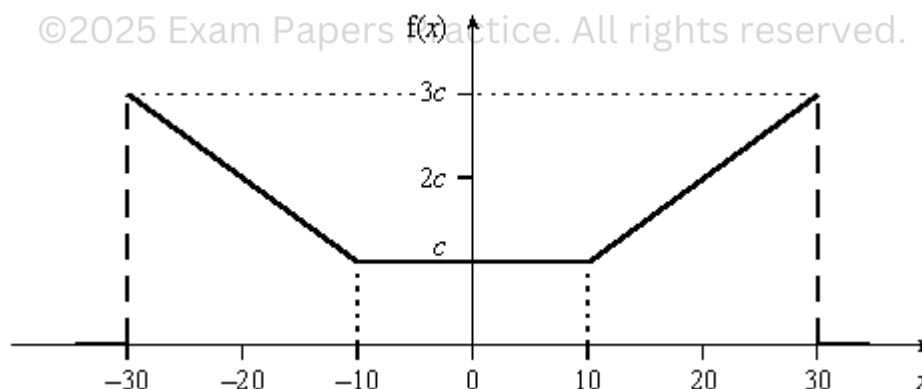
$$f(t) = \begin{cases} \frac{t^2}{18} & 0 \leq t < 3 \\ \frac{1}{4}(5-t) & 3 \leq t \leq 5 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Write down the value of $P(T = 4)$. (1)
- (b) Show that the median waiting time is 3 minutes. (2)
- (c) Find the probability that customers have to wait for less than 3.5 minutes to be served. (5)
- (d) Calculate the mean time that customers have to wait in order to be served. (5)

(Total 13 marks)

Q35.

The probability density function, $f(x)$, for the difference in millimetres from the nominal length of metal rods cut by a faulty machine, is shown by the following graph.



- (a) Show that the value of the constant c , as used on the vertical axis, is 0.01. (3)
- (b) Determine:
 - (i) $P(X > 20)$; (2)

(ii) $P(X < 10)$.

(2)

(Total 7 marks)

Q36.

All consultations with an optician are by appointment. The time, T minutes by which an appointment is delayed has the following probability density function.

$$f(t) = \begin{cases} \frac{1}{54}t^2 & 0 \leq t < 3 \\ \frac{1}{6} & 3 \leq t < 6 \\ \frac{1}{24}(10 - t) & 6 \leq t < 10 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(4)

- (b) Use your graph to find:

(i) $P(3 \leq T < 6)$;

(2)

(ii) $P(T > 6)$;

(2)

(iii) the lower quartile

(4)

- (c) What proportion of appointments is delayed by less than 2 minutes?

(4)

(Total 16 marks)

Q37.

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The queuing time, X minutes, at a post office maybe modelled by the following probability density function, where c is a constant.

$$f(x) = \begin{cases} c & 0 \leq x \leq 1 \\ \frac{c}{2}(3 - x) & 1 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f .

(4)

- (b) By considering your sketch, or otherwise, find the value of c .

(3)

- (c) Determine $P(X < 2)$.

(3)

(Total 10 marks)

Q38.

The continuous uniform random variable X has the following probability density function.

$$f(x) = \begin{cases} \frac{1}{6} & 6 \leq x \leq 12 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find values for the mean, μ , and variance, σ^2 , of X .

(2)

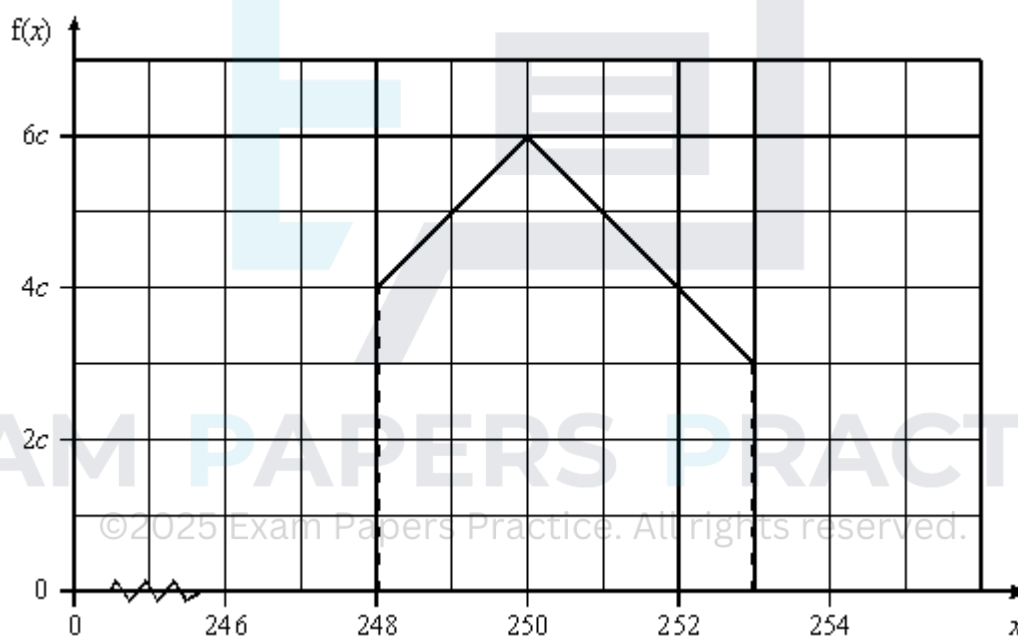
- (b) Determine $P(X > \mu + \sigma)$.

(3)

(Total 5 marks)

Q39.

The probability density function for the volume, X millilitres, of tonic water in small bottles is shown by the following graph.



- (a) Show that the value of the constant c , as used on the vertical axis, is $\frac{2}{47}$.

(4)

- (b) State the probability that a bottle contains exactly 250 ml of tonic water.

(1)

- (c) Determine the probability that a bottle contains between 249 ml and 252 ml of tonic water.

(3)

(Total 8 marks)

Q40.

The continuous random variable X has the following probability density function.

$$f(x) = \begin{cases} kx & 0 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Show that the value of the constant k is $\frac{1}{8}$. (2)
- (b) Given that $E(X) = 2\frac{2}{3}$, calculate the variance of X . (4)
- (c) Hence calculate $\text{Var}(5X - 4)$. (2)
- (d) Calculate the 90th percentile of X . (3)
- (Total 11 marks)**

Q41.

In an attempt to economise on her mobile phone costs, Tara times her calls to ensure that no call lasts longer than 6 minutes. The resultant lengths of her calls, T minutes, may be regarded as a continuous random variable with the following probability density function.

$$f(t) = \begin{cases} k(2t + 3) & 1 \leq t \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f . (3)
- (b) By considering your sketch, or otherwise, show that the value of the constant k is $\frac{1}{50}$. (3)
- (c) Determine the probability that the length of a call is:
- (i) at most 3 minutes; (3)
- (ii) between 2 and 5 minutes. (3)

(Total 12 marks)

Q42.

The total amount of added time, S minutes, played per match by a team in a local football league is distributed with probability density function given by:



$$f(s) = \begin{cases} ks^2 & 1 \leq s \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Show that the exact value of the constant k is $\frac{3}{124}$. (4)
- (b) Determine $P(S < 3)$. (3)
- (c) Calculate the median amount of added time played per match, giving your answer to the nearest minute. (4)
- (Total 11 marks)

Q43.

The time, H hours, to carry out a 24 000 miles service on a car may be regarded as a continuous random variable with the following probability density function.

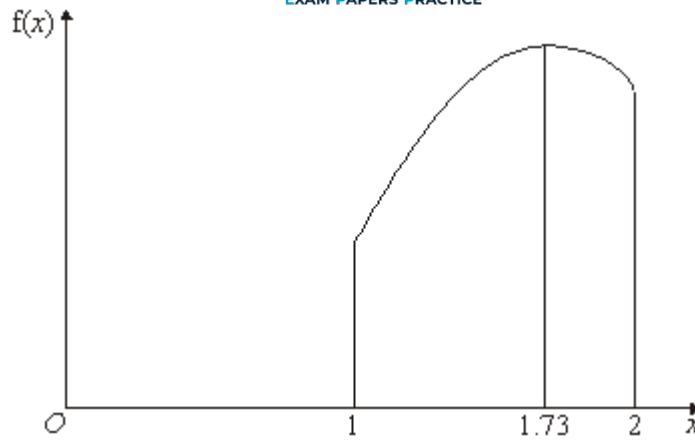
$$f(h) = \begin{cases} c(7 - 2h) & 1 \leq h \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f . (3)
- (b) By considering your sketch, or otherwise, show that the value of the constant c is $\frac{1}{6}$. (3)
- (c) Calculate the probability that a service takes more than 2 hours. (2)
- (d) Given that a service is **not** completed in 2 hours, calculate the probability that it is completed in less than $2\frac{1}{2}$ hours. (5)
- (Total 13 marks)

Q44.

The sketch below shows the probability density function $f(x)$ of the continuous random variable X .

($f(x) = 0$ outside the interval $1 \leq x \leq 2$.)



You are given that $\int_1^{1.56} f(x) \, dx = 0.5$ $\int_{1.73}^2 f(x) \, dx = 0.291$

$$\int_1^2 x f(x) \, dx = 1.54 \qquad \int_1^2 (x - 1.54)^2 f(x) \, dx = 0.0751$$

(a) Write down the value of:

(i) $\int_1^2 f(x) \, dx$;

(ii) the mean of X ;

(iii) the median of X ;

(iv) the mode of X ;

(v) the range of X .

(b) Find:

(i) the probability that X lies between the median and the mode;

(2)

(ii) the standard deviation of X ;

(2)

(iii) $E(X^2)$.

(2)

(Total 11 marks)

Q45.

The random variable X has probability density function:

$$f(x) = \begin{cases} px+q & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

where p and q are constants.

- (a) Show that $\frac{1}{2}p + q = 1$. (3)
- (b) Find the mean of X in terms of p and q . (3)
- (c) The mean of X is 0.6. Show that $p = 1.2$ and find the value of q . (3)
- (d) Find the standard deviation of X . (4)
- (Total 13 marks)

Q46.

The random variable, X , has probability density function

$$f(x) = \begin{cases} cx + d & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases} \quad \text{where } c \text{ and } d \text{ are constants}$$

- (a) Show that $2c + 2d = 1$. (3)
- (b) Find the median of X :
- (i) when $c = 0$ and $d = 0.5$; (2)
- (ii) when $c = 0.5$ and $d = 0$. (3)
- (c) The following pairs of values are suggested for c and d .

Suggestion A: $c = 0.5$ and $d = 0.5$

Suggestion B: $c = -0.5$ and $d = 1$

Suggestion C: $c = 1.5$ and $d = -1$

Only one of the suggestions provides a valid probability density function.

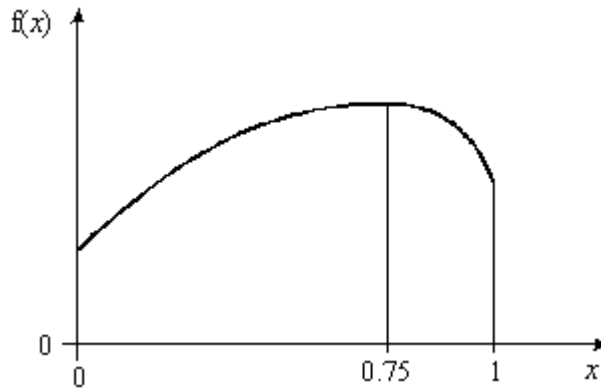
For **each** suggestion state whether or not it is valid. If you state it is not valid, give a reason.

(5)
(Total 13 marks)

Q47.

The sketch below shows the probability density function $f(x)$ of the continuous random variable X .

$[f(x) = 0 \text{ outside the interval } 0 \leq x \leq 1]$



- (a) Explain why the value of the mode of X is 0.75. (1)
- (b) You are given that:
- $E(X) = 0.542$ $E(X^2) = 0.369$
- $\int_0^{0.557} f(x)dx = 0.500$ $\int_0^{0.542} f(x)dx = 0.483$ $\int_0^{0.75} f(x)dx = 0.719$
- (i) Find the standard deviation of X . (3)
- (ii) Find the probability that X exceeds the mode. (2)
- (iii) Write down the value of the median of X . (1)
- (iv) Evaluate the probability that X lies between the median and the mean. (3)
- (Total 10 marks)**

Q48.

The errors, in grams, made by a butcher's scales may be modelled by a random variable, y , with probability density function

$$f(y) = \begin{cases} k & -6 < y < 14 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Show that $k = 0.05$. (2)
- (b) Write down the mean value of Y . (1)
- (c) (i) Use integration to find $E(Y^2)$. (3)

- (ii) Hence, or otherwise, find the standard deviation of Y . (3)
- (d) (i) Find the probability that an error is negative. (1)
- (ii) Show that the probability of an error having magnitude less than 2 is 0.2. (2)
- (iii) Find the probability that the magnitude of an error is greater than 4. (2)
- (Total 14 marks)**



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